SUMS OF TWO SQUARES AND HEEGNER NORMS WITH FEW PRIME FACTORS, IV: THE SCALLOPED PARSEVAL FORMULA, THE HEEGNER CLASSIFICATION, AND EISENSTEIN DISPERSION

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Abstract. Parts I–III studied the cusp form $G_{m,m'}$ attached to $\theta^2(mz)\theta^2(m'z)$, whose Fourier coefficients are the remainders of the sieve for $r(n)r(N-n)$. This part contains three results. (1) The Petersson norm of G is given *exactly* by a Parseval sum over the cusps whose lower boundary is the scalloped curve $y=\sqrt{1-\{x\}^2}$ of Waibel's fundamental domain, with correlation terms $a(k)\overline{a(k+jQ)}$ at distances multiple of the width. Evaluated to $p=4\cdot 10^5$, it reproduces the Rankin–Selberg values where these have converged and determines the coefficient of $\log p$ in $\|G_{1,p}\|^2=\frac{1}{\pi}(\log p)^2+B\log p+O(1)$ in closed form: $B=0.48204$ is the sum of 0.383 from the free part of the two main cusps, $2\kappa/\pi=0.0288$ from the region between the arcs and the line $y=1$, $\kappa=\int_{-1/2}^{1/2}\log(1-x^2)^{-1/2}\mathrm{d}x$ (both by Mellin inversion), and 0.0702 from the coefficients $k\geq p$, a sum of exponential integrals $E_1(\pi\beta)$, $E_1(4\pi\beta)$ weighted by $r(\beta)^2$; the Siegel main term contributes nothing to B. (2) For the seven odd discriminants of class number one, $D\in\{3,7,11,19,43,67,163\}$, the theta series θ_D of the norm form is the Eisenstein series $E_1(z,\chi_{-D})$ and $\Theta_{1,p}^{(D)}=\theta_D(z)\theta_D(pz)$ has the square-free level Dp, all of whose cusps are Atkin–Lehner. We prove, uniformly in D, the Siegel main term with closed-form densities at D and p; the signs $\Theta|W_p=\chi(p)\Theta$, $\Theta|W_D=-\chi(p)\Theta$; the spectral support on the newforms with $a_\varphi(D)=\chi_{-D}(p)$, $a_\varphi(p)=-\chi_{-D}(p)$ (no inert vanishing); the Rankin–Selberg formula with the constant $C_D=w_D^2\sqrt{D}/(8\pi^2)$; and the lower bound $\|G_{1,p}^{(D)}\|^2\geq\frac{6}{\pi\psi(D)}(\log p)^2+O(\log p)=\frac{2}{\mathrm{vol}(\Gamma_0(D)\backslash\mathbb{H})}(\log p)^2+O(\log p)$, which holds for all nine discriminants and is the asymptotic for $D=4$. Everything is verified numerically for $D\leq 43$ (79 newforms, Rankin–Selberg ratios within 0.5%) and the norm scans for all seven odd D. (3) For the Eisenstein integers the dispersion form of Part III is rederived: the Fourier transform of the circle $x^2+xy+y^2\equiv a$ modulo any c prime to 3 is a classical Kloosterman sum with the hexagonal dual norm, the ramified prime 3 takes over the role that 2 plays for the Gaussian integers, and the obstruction to a level beyond 1/2 is unchanged. In the course of this we correct an error of Parts II–III: the Kloosterman sums of the dispersion form are $S(\overline{16m}N,|h|^2;m'k)$, not $S(16N,m|h|^2;m'k)$.

1. Introduction and statement of results

1.1. **Setting.** Parts I–III [2, 3, 4] prove the level of distribution 1/2 for the weighted sequence $r(n)r(N-n)$, r the number of representations as a sum of two squares, and with it $\Omega(n(N-n))\leq 7$, $\max(\Omega(n),\Omega(N-n))\leq 5$ for every large $N\equiv 2 \pmod 4$, through the cusp forms $G_{m,m'}\in S_2(\Gamma_0(4mm'))$ with $\theta^2(mz)\theta^2(m'z)=E_{m,m'}+G_{m,m'}$: the Fourier coefficient $a_{G_{m,m'}}(N)=T(m,m';N)-M(m,m';N)$ is the remainder of the sieve at the modulus pair (m,m'), and the uniform bound $|a_G(N)|\ll N^{1/2+\varepsilon}\|G\|$ (Part I, Theorem 2.3) with $\|G_{m,m'}\|^2\asymp(\log mm')^2$ (Part II, Theorems 1.5 and 6.14) gives the level. Part III located the obstacles to a level beyond 1/2 and to the pair $(7,4)$. The present part answers three questions that arose from it: how large exactly is the norm (the term $B\log p$ which Part II, Remark 6.16, could only estimate);

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what happens for the other imaginary quadratic fields of class number one; and whether the Eisenstein integers, with their square-free level, change the dispersion problem.

1.2. **Results.** Let $G \in S_2(\Gamma_0(L))$. Waibel's formula [11, (22)] (Part I, Lemma 2.8) writes $\|G\|^2$ as a sum over the cusp classes of integrals of $|G|_2\varrho|^2$ over the union of the L translates $t\mathcal{F}$, $0 \le t < L$, of the standard fundamental domain, whose lower boundary is the *scalloped* curve $y = s(x) = \sqrt{1-\{x\}^2}$ ($\{x\}$ the distance to the nearest integer), not a horizontal line. Part II bounded these integrals by the strips $y \ge 1$ and $y \ge \sqrt{3}/2$. Parseval over the exact region gives an identity.

Theorem 1.1 (Scalloped Parseval)**.** *For $G|_2\varrho = \sum_{k\ge1} b(k)e(kz/Q)$ at a cusp of width Q, with $\Psi_j(c) = \int_{-1/2}^{1/2} \cos(2\pi jx)e^{-c\sqrt{1-x^2}}\mathrm{d}x$,*

$$\|G\|^2 = \sum_{d\mid L} w_d \sum_{k\ge1}\sum_{j} b_d(k)\overline{b_d(k+jQ_d)}\,\frac{\Psi_j(c_{k,j})}{c_{k,j}}, \qquad c_{k,j} = \frac{2\pi(2k+jQ_d)}{Q_d},$$

*with explicit weights w_d (Proposition 2.1). For $G = G_{1,p}$ the formula, evaluated with $k \le 14p$, reproduces the Rankin–Selberg estimates of Part II where these have converged (*7.42858 *against* 7.42871 *at $p = 101$), lies inside the sandwich $\Lambda_1 \le \|G\|^2 \le \Lambda_{\sqrt{3}/2}$ at every p, and gives, over $p \ge 10^4$,*

$$\|G_{1,p}\|^2 = \frac{1}{\pi}(\log p)^2 + B\log p + O(1), \qquad B = 0.48204\ldots,$$

where B is the explicit constant of Theorem 2.4: 0.383 *is the free part $\frac{1}{4\pi}(S(p/4\pi)+S(p/\pi))$ of the two main cusps (Part II, Lemma 6.3),* 0.0288 $= 2\kappa/\pi$ *the arcs acting on that free part, $\kappa = \int_{-1/2}^{1/2}\log(1-x^2)^{-1/2}\mathrm{d}x =$* 0.045231 *(Proposition 2.2),* 0.0569 *and* 0.0119 *the coefficients $k \ge p$ of the two main cusps at height* 1 *and through the arcs, $\frac{1}{\pi}\sum_\beta r(\beta)^2(E_1(\pi\beta)+E_1(4\pi\beta))$ and its arc version,* 0.0019 *the cusp of denominator* 2 *and* -0.0005 *the correlation terms; the Siegel main term and all cross terms are $O(1)$. The fit over $10^4 \le p \le 4\cdot10^5$ gives* 0.4817*, the local slopes at the largest primes* 0.4819 *and* 0.4820*.*

The value $B \approx 0.46$ of Part II, Remark 6.16, was a fit to Rankin–Selberg estimates that had not converged; the fit of the exact formula gives 0.48, and Theorem 2.4 the constant itself.

Let now $D \in \{3,7,11,19,43,67,163\}$ be an odd discriminant of class number one, $\chi = \chi_{-D}$, w_D the number of units, $N_D(x,y) = x^2+xy+\frac{D+1}{4}y^2$, $r_D(n) = w_D\sum_{d\mid n}\chi(d)$, $\theta_D = \sum r_D(n)e(nz)$ and, for a prime $p \ne D$,

$$\Theta_{1,p}^{(D)} = \theta_D(z)\theta_D(pz) = E + G_{1,p}^{(D)} \in M_2(\Gamma_0(Dp)),$$
$$a_G(k) = T(k) - M(k) = \sum_{\alpha+p\beta=k} r_D(\alpha)r_D(\beta) - M(k).$$

Theorem 1.2 (The odd Heegner discriminants)**.** (i) *$M(k) = \frac{4\pi^2k}{Dp}\prod_\ell \beta_\ell(k)$ with the standard densities at $\ell \nmid Dp$ and, with $v = v_\ell(k)$ and $\varepsilon = -\chi_{-D}(p)$,*

$$\beta_p(k) = (1+\chi(p)) - \chi(p)\Big(1+\frac{1}{p}\Big)p^{-v}, \qquad \beta_D(k) = (1+\varepsilon) - \varepsilon\Big(1+\frac{1}{D}\Big)D^{-v}.$$

(ii) $\Theta|_2W_p = \chi(p)\Theta$, $\Theta|_2W_D = -\chi(p)\Theta$, $\Theta|_2W_{Dp} = -\Theta$.

(iii) *A newform φ of level Dp occurs in $G_{1,p}^{(D)}$ only if $a_\varphi(D) = \chi_{-D}(p)$ and $a_\varphi(p) = -\chi_{-D}(p)$ (root number $+1$); an oldform from level p only if $a_f(p) = -\chi_{-D}(p)$; an oldform from level D only if $\chi_{-D}(p) = +1$. There is no inert vanishing: for every p both signs of $\chi(p)$ admit newforms.*

(iv) $\langle \Theta_{1,p}^{(D)}, \varphi\rangle = C_D\,\chi(p)\,\eta_p(\varphi)\,L(1,\varphi)L(1,\varphi\otimes\chi_{-D})$ *with* $C_D = \dfrac{w_D}{4\pi L(1,\chi_{-D})} = \dfrac{w_D^2\sqrt{D}}{8\pi^2}$.

(v) $\|G_{1,p}^{(D)}\|^2 \;\geq\; \dfrac{6}{\pi(D+1)}(\log p)^2 + b_D\log p + O(1)$ *with an explicit* b_D*; and for all nine discriminants, including* $D = 4, 8$*, the two wide Atkin–Lehner cusps give, with* $\psi(D) = D\prod_{\ell\mid D}(1+1/\ell)$*,*

$$\|G_{1,p}^{(D)}\|^2 \;\geq\; \frac{6}{\pi\,\psi(D)}(\log p)^2 + O(\log p) = \frac{2}{\mathrm{vol}(\Gamma_0(D)\backslash\mathbb{H})}(\log p)^2 + O(\log p),$$

which is the asymptotic for $D = 4$ *(Part II, Theorem 6.14).*

The constants are those of the analytic class number formula with $h = 1$, $w_D^2 L(1,\chi_{-D})^2 = 4\pi^2/D$, which makes them independent of w_D; the remaining factor $\prod_{\ell\mid D}(1+1/\ell)^{-1}$ is the Euler factor at the ramified primes, where $\sigma_\chi(\ell^j) = 1$. Section 4 verifies (i)–(iv) for $D \leq 43$ (79 newforms, Rankin–Selberg ratios in $[0.9936, 1.0048]$) and (v) for all seven odd D (fitted coefficients 0.4815, 0.2408, 0.1601, 0.0961, 0.0443, 0.0266, 0.0128 against 0.4775, 0.2387, 0.1592, 0.0955, 0.0434, 0.0281, 0.0116).

Theorem 1.3 (Eisenstein dispersion)**.** *For* $(c,3) = 1$*,* c *odd or even,* $(a,c) = 1$ *and* $h \in \mathbb{Z}^2$*, with* $|h|_3^2 = h_1^2 - h_1h_2 + h_2^2$ *the hexagonal dual norm,*

$$\sum_{\substack{x,y \bmod c\\ x^2+xy+y^2\equiv a\ (c)}} e\Big(\frac{h_1x+h_2y}{c}\Big) = \sum_{g\mid(c,h)} g\,\chi_{-3}\Big(\Big(\frac{c}{g}\Big)^*\Big) S\Big(a,\ \overline{3}\Big|\frac{h}{g}\Big|_3^2;\ \frac{c}{g}\Big).$$

The remainder $a_{G^{(3)}_{m,m'}}(N) - a_{G^{(3)}_{m,3m'}}(N)$ *of the sieve for* $r_3(n)r_3(N-n)$ *has as its leading term a trilinear form in the sums* $S(\overline{27m}N, |h|_3^2; m'k)$ *over all moduli* $m'k$ *prime to* 3*, with a dual condition* $3 \mid |h|_3^2$ *in place of* $2 \mid |h|^2$*; its bilinear form in* (m',k) *and the Cauchy–Schwarz and Poisson sizes are those of Part III, and the obstruction to a level beyond* $1/2$ *is the same.*

1.3. **A correction to Parts II–III.** Part III, Section 3.2 (and Part II, Section 8.1) rewrote $S(N\bar m, \overline{16}|h|^2; c)$ as $S(16N, m|h|^2; c)$. This is false: a Kloosterman sum depends on its two arguments only through their product modulo the modulus, $S(a,b;c) = S(1,ab;c)$, and the products differ by $(16m)^2$. The correct form is $S(\overline{16m}N, |h|^2; c)$, the inverse of $16m$ taken modulo the modulus, and it cannot be removed. The Weil, Cauchy–Schwarz and Poisson computations of Part III, which only open the sums, are unaffected; the Kuznetsov heuristic of its Section 3.5(b), which needs fixed integer arguments, is withdrawn (the integer representatives $(N+tc)/(16m)$ of the first argument vary with c); the correlation experiment of its Section 4.4 has been rerun with the correct sums, with the same conclusion. The corrected versions of Parts II and III are the ones cited here.

2. The scalloped Parseval formula and the anatomy of B

Throughout, $G \in S_2(\Gamma_0(L))$, and for a class of pairs (a,c) modulo L with $(c,L) = d$ let $\varrho = \varrho_{a,c} \in \mathrm{SL}_2(\mathbb{Z})$ have first column (a,c), $\hat d \in \{d, d/2\}$ as in Part I, Lemma 2.9, $Q = L/\hat d$, and $G|_2\varrho = \sum_{k\geq1} b(k)e(kz/Q)$. Part I, Lemma 2.8 is

$$\|G\|^2 = \frac{\mu}{\tilde\mu}\sum_{c\ (L)}\sum_{\substack{a\ (L)\\(a,c,L)=1}} I(\varrho_{a,c}), \qquad I(\varrho) = \int_{\bigcup_{0\leq t<L} t\mathcal{F}} |G|_2\varrho|^2\,\mathrm{d}x\,\mathrm{d}y, \tag{1}$$

and $\bigcup_{0\leq t<L} t\mathcal{F} = \{-\frac12 \leq x \leq L - \frac12,\ y \geq s(x)\}$ with $s(x) = \sqrt{1-\{x\}^2}$ of period 1.

Proposition 2.1. *For $j \in \mathbb{Z}$ and $c > 0$ let $\Psi_j(c) = \int_{-1/2}^{1/2} e(-jx)\, e^{-c\sqrt{1-x^2}}\,\mathrm{d}x$ (real, $\Psi_{-j} = \Psi_j$). Then*

$$I(\varrho) = L \sum_{k\ge 1} \sum_{j \ge -k/Q} b(k)\overline{b(k+jQ)}\,\frac{\Psi_j(c_{k,j})}{c_{k,j}}, \qquad c_{k,j} = \frac{2\pi(2k+jQ)}{Q},$$

and therefore $\|G\|^2 = \sum_{d\mid L} w_d \sum_{k,j} b_d(k)\overline{b_d(k+jQ_d)}\Psi_j(c_{k,j})/c_{k,j}$ with $w_d = Q_d$ if $\hat{d} = d$ and $w_d = Q_d/4$ if $\hat{d} = d/2$. The terms $j = 0$ with $\Psi_0(c)$ replaced by e^{-c} are the height-1 sums $\Lambda_1(d)$ of Part II; $\Psi_0(c) - e^{-c} > 0$ integrates over the region between the arcs and the line $y = 1$; the terms $j \ne 0$ are correlations of coefficients at distance a multiple of the width, which a flat bottom annihilates. Since $c_{k,j} \ge 2\pi j$ and $\Psi_j(c) \le e^{-c\sqrt{3}/2}$, only $j \le 7$ contribute above 10^{-16}.

Proof. $|G|_2\varrho|^2 = \sum_{k,k'} b(k)\overline{b(k')}e((k-k')x/Q)e^{-2\pi(k+k')y/Q}$. Integrating in y from $s(x)$ to ∞ gives $e^{-c\,s(x)}/c$ with $c = 2\pi(k+k')/Q$, and since s has period 1 and the x-range consists of L consecutive unit periods,

$$\int_{-1/2}^{L-1/2} e\Big(\frac{(k-k')x}{Q}\Big)e^{-c\,s(x)}\mathrm{d}x = \Big(\sum_{t=0}^{L-1} e\Big(\frac{(k-k')t}{Q}\Big)\Big)\int_{-1/2}^{1/2} e\Big(\frac{(k-k')x}{Q}\Big)e^{-c\,s(x)}\mathrm{d}x,$$

where the geometric sum is L if $Q \mid k-k'$ and 0 otherwise. With $k' = k + jQ$ the inner integral is $\Psi_j(c)$. The count of pairs in a class and $\mu/\tilde{\mu} = 1/(L\varphi(L))$ are as in Part II, Lemmas 6.2 and 6.12. □

For $G = G_{1,p}$, $L = 4p$, the six classes are $d \in \{1, 4, 2\}$ (widths $4p, p, 4p$; $b = a_{G_{1,p}}/Q$ at the Atkin–Lehner cusps, $b = \frac{1}{p}(T^{\mathrm{odd}} - M^{\mathrm{odd}})$ at the cusp of denominator 2) and $d \in \{p, 4p, 2p\}$ (widths $4, 1, 4$). The script `subleading_log_anatomy.py` evaluates the formula with $k \le 14p$ and $j \le 8$, Ψ_j by Gauss–Legendre quadrature on a logarithmic grid in c with cubic splines.

Proposition 2.2 (The arc term)**.** *At a cusp of width Q put $Y = Q/4\pi$ and $\kappa = \int_{-1/2}^{1/2} \log(1-x^2)^{-1/2}\mathrm{d}x = 0.045231$. Then*

$$\sum_{k\ge 1} \frac{r(k)^2}{k}\Psi_0(k/Y) = 2(\log Y)^2 + (c_1 + 4\kappa)\log Y + O(1),$$

with $c_1 = 9.758$ the constant of Part II, Lemma 6.3; so the arcs add $4\kappa \log Y$ to the free part at each of the two main cusps, i.e. $\frac{2\kappa}{\pi}\log p = 0.028795 \log p$ in total.

Proof. $\int_0^\infty \Psi_0(c)c^{w-1}\mathrm{d}c = \Gamma(w)\int_{-1/2}^{1/2}(1-x^2)^{-w/2}\mathrm{d}x = \Gamma(w)(1 + \kappa w + O(w^2))$, and Mellin inversion with $D(1+w) = \sum r(k)^2 k^{-1-w} = 4w^{-2}(1 + a_1 w + \dots)$ gives the residue at $w = 0$ as in Part II, Lemma 6.3, with a_1 replaced by $a_1 + \kappa$. The antiderivative of $\log(1-x^2)$ is $x\log(1-x^2) - 2x + \log\frac{1+x}{1-x}$, whence the value of κ. (The first-order approximation $\Psi_0(c) - e^{-c} \approx c\kappa_1 e^{-c}$, $\kappa_1 = \int(1-\sqrt{1-x^2})\mathrm{d}x = 0.04339$, gives $4\kappa_1$ instead: the higher orders in c also contribute to the $\log Y$ term.) □

2.1. **The anatomy of B.** Write $\|G_{1,p}\|^2 - \frac{1}{\pi}(\log p)^2 = C_1' + C_2 + C_3 + C_4$ with $C_1 = \frac{1}{4\pi}(S(p/4\pi) + S(p/\pi))$ the free part of the two main cusps, $C_1' = C_1 - \frac{1}{\pi}(\log p)^2 = 0.38301 \log p + O(1)$, $C_2 = \Lambda_1(\text{main cusps}) - C_1$ (the correction $r(k)^2 \to |a(k)|^2$: the Siegel main term for $k < p$, the terms with $\beta \ge 1$ for $k \ge p$), $C_3 = \Lambda_1$ of the other four cusps, and $C_4 = \|G\|^2 - \Lambda_1$ (the part below height 1). Table 1 and Figure 1 give the result of the exact formula for twelve primes from 101 to 400009 (a few seconds per prime).

So $B \approx 0.48$: 0.383 and 0.029 are proved by Mellin inversion, and the remaining 0.07 is the Siegel and tail corrections at height 1 (0.057) and through the arcs (0.012), plus the other cusps

p	C_1'	C_2	C_3	C_4 arc, main	free part	C_4 corr.	total, $/\log p$
101	1.128	−0.815	0.154	0.164	0.151	−0.001	0.649, 0.141
1009	1.999	−0.711	0.162	0.259	0.218	−0.002	1.727, 0.250
10007	2.876	−0.582	0.166	0.352	0.284	−0.003	2.830, 0.307
100003	3.758	−0.453	0.169	0.446	0.350	−0.004	3.939, 0.342
400009	4.289	−0.374	0.171	0.502	0.390	−0.005	4.607, 0.357
fit b ($p \ge 10^4$)	0.3829	0.0566	0.0020	0.0407	0.0288	−0.0005	0.4817
local slope, $p \ge 10^5$	0.3830	0.0568	0.0019	0.0407	0.0288	−0.0005	0.4819
proved (Theorem 2.4)	0.3830	0.0569	0.0019	0.0407	0.0288	−0.0005	0.4820

TABLE 1. The parts of $\|G_{1,p}\|^2 - \frac{1}{\pi}(\log p)^2$; C_3 includes the arc terms of the four narrow cusps. The fit of the total is the three-parameter fit of the exact norm, with $a = 0.31845$ for $(\log p)^2$ ($1/\pi = 0.31831$) and $c = -1.607$; the local slopes are $(y(p_2) - y(p_1))/\log(p_2/p_1)$ over the last three primes. The part below height 1 is $0.041 \log p$, against the bound $0.134 \log p$ of Part II, Section 6.7. Scripts `subleading_log_anatomy.py`, `closed_form_B.py`; floating point.

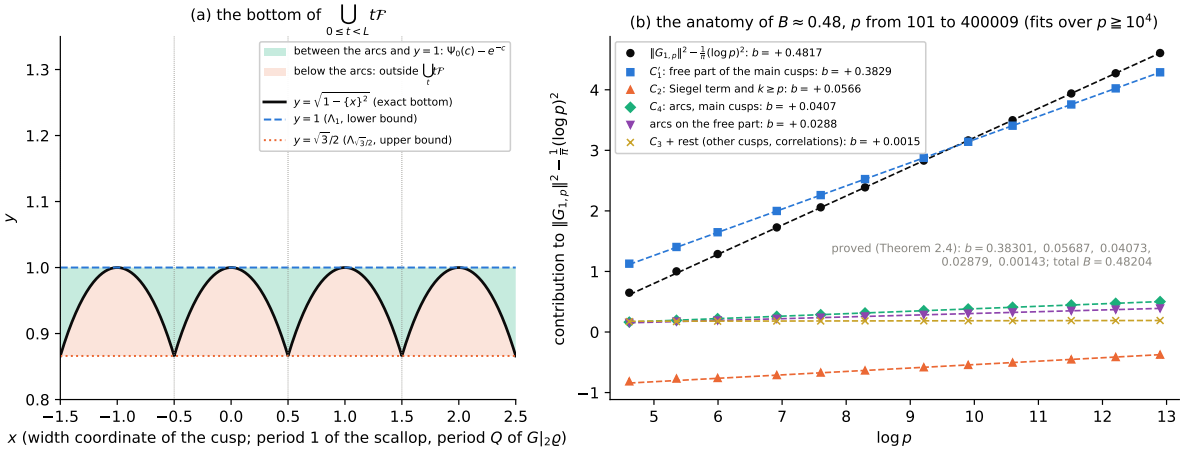


FIGURE 1. (a) The lower boundary of $\bigcup_{0 \le t < L} t\mathcal{F}$ in the width coordinate of a cusp: the arcs $y = \sqrt{1 - \{x\}^2}$, the line $y = 1$ of the lower bound Λ_1 and the line $y = \sqrt{3}/2$ of the upper bound $\Lambda_{\sqrt{3}/2}$ of Part II; the shaded region between the arcs and $y = 1$ is what $\Psi_0(c) - e^{-c}$ integrates. (b) The parts of $\|G_{1,p}\|^2 - \frac{1}{\pi}(\log p)^2$ against $\log p$ for the twelve primes of Table 1, with their least-squares lines over $p \ge 10^4$; the proved slopes of Theorem 2.4 are listed. Script `plot_scalloped_anatomy.py`; floating point.

(0.002) and the correlations (−0.0005). These are not Mellin transforms of Dirichlet series, and the next subsection evaluates them by a different mechanism.

2.2. **The closed form of B.** For $k \ge p$ the coefficient is $a(k) = \sum_{\beta \ge 0} r(\beta) r(k - p\beta) - M(k)$ with $r(0) = 1$: the term $\beta = 0$ is the free part $r(k)$, and $T'(k) = \sum_{\beta \ge 1} r(\beta) r(k - p\beta)$ is the tail. In $|a(k)|^2$ the squares $r(k - p\beta)^2$, summed over k, produce a logarithm by $\sum_{a \le X} r(a)^2 \sim 4X \log X$; every other product is a shifted convolution of r, or involves the Siegel main term, and is bounded by the following elementary estimates.

Lemma 2.3. *Uniformly for $X \geq 1$ and integers $h \geq 0$:*
(i) *for $h \geq 1$, $\sum_{a \leq X} r(a)r(a+h) \ll (X+h)\,\sigma_{-1}(h) + \sqrt{X+h}\,d(h)$;*
(ii) *$\sum_{a \leq X} r(a)\sigma_{-1}(a+h) \ll X+h$ and $\sum_{a \leq X} \sigma_{-1}(a)\sigma_{-1}(a+h) \ll X+h$;*
(iii) *$\sum_{a \leq X} r(a)^2 = 4X\log X + O(X)$ and $\sum_{a \leq X,\ a \equiv 2\,(4)} r(a)^2 = X\log X + O(X)$.*

Proof. (i) The sum counts $(x,y,z,w) \in \mathbb{Z}^4$ with $x^2+y^2 \leq X$ and $z^2+w^2-x^2-y^2 = h$. The substitution $u_1 = z-x$, $v_1 = z+x$, $u_2 = w-y$, $v_2 = w+y$ is injective, gives $u_1v_1+u_2v_2 = h$ and keeps $|u_i|, |v_i| \leq U := 2\sqrt{X+h}$. If $u_1 = 0$, then $u_2v_2 = h$ has at most $2d(h)$ solutions and v_1 at most $2U+1$ values; symmetrically for $u_2 = 0$; together $\ll Ud(h)$. If $u_1u_2 \neq 0$ and $g = (u_1,u_2)$, there is no solution unless $g \mid h$, and then (v_1,v_2) runs over a coset of $\mathbb{Z}\,(u_2/g, -u_1/g)$, which has at most $1+2Ug/\max(|u_1|,|u_2|)$ points in the box. Summing over (u_1,u_2), the 1 gives $\ll U^2$, and with $u_i = gu_i'$, $(u_1',u_2') = 1$, the second term gives $2U\sum_{g\mid h}\sum_{|u_i'| \leq U/g} 1/\max(|u_1'|,|u_2'|) \ll U\sum_{g\mid h} U/g = U^2\sigma_{-1}(h)$, as at most $8m$ pairs have $\max(|u_1'|,|u_2'|) = m$.

(ii) Write $\sigma_{-1}(a+h) = \sum_{e\mid a+h} 1/e$ and split at $E = \log^2(2X+2h)$. The divisors $e > E$ contribute at most $E^{-1}\sum_{a\leq X} r(a)d(a+h) \leq E^{-1}\big(\sum_{a\leq X} r(a)^2\big)^{1/2}\big(\sum_{n\leq X+h} d(n)^2\big)^{1/2} \ll E^{-1}(X+h)\log^2(2X+2h) \ll X+h$. For $e \leq E$, $\sum_{a\leq X,\ e\mid a+h} r(a)$ counts the lattice points of the disc $x^2+y^2 \leq X$ in the $N_2(-h;e)$ residue classes modulo e with $x^2+y^2 \equiv -h$, so it is $N_2(-h;e)\big(\pi X/e^2 + O(\sqrt{X}/e+1)\big)$ with $N_2(t;e) \leq e\,d(e)^2$ (Part II, proof of Lemma 6.4), and the sum over $e \leq E$ is $\ll X\sum_e d(e)^2e^{-2} + \sqrt{X}\sum_{e\leq E} d(e)^2e^{-1} + \sum_{e\leq E} d(e)^2 \ll X+h$. The second bound is the same argument with $r(a)$ replaced by $\sigma_{-1}(a) = \sum_{f\mid a} 1/f$: for $e,f \leq E$ the number of $a \leq X$ with $f \mid a$, $e \mid a+h$ is at most $X(e,f)/(ef)+1$ and $\sum_{e,f}(e,f)(ef)^{-2}$ converges, while $e > E$ or $f > E$ contribute $\ll E^{-1}\sum_{a\leq X}\big(d(a)\sigma_{-1}(a+h) + \sigma_{-1}(a)d(a+h)\big) \ll E^{-1}(X+h)\log^2(2X+2h)$ by Cauchy–Schwarz.

(iii) The first statement is Part II, Lemma 6.3. For the second, $r(2m) = r(m)$ gives $\sum_{a\leq X,\ a\ \text{odd}} r(a)^2 = R(X) - R(X/2)$ with $R(X) = \sum_{a\leq X} r(a)^2$, and $a \equiv 2 \pmod 4$ means $a = 2m$ with m odd, so the sum is $R(X/2) - R(X/4) = X\log X + O(X)$. □

Theorem 2.4 (The closed form of B)**.** *Let $E_1(x) = \int_x^\infty e^{-t}\,\mathrm{d}t/t$, $s = \sqrt{1-x^2}$, $I_j(v_0;\lambda) = \int_{v_0}^\infty \Psi_j(\lambda v)\,\mathrm{d}v/(\lambda v)$, and $r(0) = 1$. Then*

$$\|G_{1,p}\|^2 = \frac{1}{\pi}(\log p)^2 + B\log p + O(1), \qquad B = b_1+b_2+b_3+b_4+b_5+b_6 = 0.48204\ldots,$$

$$b_1 = \frac{c_1 - 2\log(4\pi^2)}{2\pi} = 0.383005, \qquad b_2 = \frac{2\kappa}{\pi} = 0.028794,$$

$$b_3 = \frac{1}{\pi}\sum_{\beta\geq 1} r(\beta)^2\big(E_1(\pi\beta) + E_1(4\pi\beta)\big) = 0.056874,$$

$$b_4 = \frac{1}{\pi}\sum_{\beta\geq 1} r(\beta)^2\int_{-1/2}^{1/2}\big(E_1(\pi\beta s) + E_1(4\pi\beta s) - E_1(\pi\beta) - E_1(4\pi\beta)\big)\mathrm{d}x = 0.011936,$$

$$b_5 = \frac{1}{\pi}\sum_{\beta\equiv 2\,(4)} r(\beta)^2\int_{-1/2}^{1/2} E_1(\pi\beta s)\,\mathrm{d}x = 0.001893,$$

$$b_6 = \sum_{j\geq 1}\Big[4\sum_{\beta\geq 0} r(\beta)r(\beta+j)\,I_j(2\beta+j;2\pi) + 2\sum_{\beta\geq 0} r(\beta)r(\beta+4j)\,I_j(\beta+2j;\pi) + 2\sum_{\beta\equiv 2\,(4)} r(\beta)r(\beta+4j)\,I_j(\beta+2j;\pi)\Big] = -0.000460.$$

Here b_1 is the free part of the two main cusps above height 1 and b_2 the arcs on it (Section 2); b_3 and b_4 are the coefficients $k \ge p$ of the two main cusps above height 1 and through the arcs; b_5 is the cusp of denominator 2 (width $4p$) with its arcs; b_6 the correlation terms. The Siegel main term, all cross terms and the three narrow cusps contribute $O(1)$.

Proof. We go through the six classes of Proposition 2.1; $O(1)$ means bounded independently of p, and $M(k) \ll k\sigma_{-1}(k)/p$ by Part II, Lemma 6.4 (the same for M^{odd}).

Main cusps, height 1. At the width $Q \in \{p, 4p\}$ the class is $\frac{1}{4\pi}\sum_k |a(k)|^2 e^{-k/Y}/k$, $Y = Q/4\pi$, so $pb/Y = 4\pi pb/Q \in \{4\pi b, \pi b\} \ge \pi b$. With $a = r + T' - M$, $|a|^2 - r^2 = 2rT' + T'^2 - 2rM - 2T'M + M^2$. By partial summation and Lemma 2.3(ii), $\sum_k r(k)M(k)e^{-k/Y}/k \ll p^{-1}\sum_k r(k)\sigma_{-1}(k)e^{-k/Y} \ll Y/p \ll 1$ and $\sum_k M(k)^2 e^{-k/Y}/k \ll p^{-2}\sum_k k\sigma_{-1}(k)^2 e^{-k/Y} \ll Y^2/p^2 \ll 1$. Next, $\sum_k r(k)T'(k)e^{-k/Y}/k = \sum_{\beta\ge1} r(\beta)\sum_{a\ge0} r(a)r(a+p\beta)e^{-(a+p\beta)/Y}/(a+p\beta)$, and by (i) and partial summation against $e^{-t/Y}/t$ the inner sum is $\ll d(p\beta)e^{-p\beta/2Y} \ll d(\beta)e^{-\pi\beta/2}$, which sums to $O(1)$; similarly $\sum_k T'(k)M(k)e^{-k/Y}/k \ll p^{-1}\sum_\beta r(\beta)\sum_a r(a)\sigma_{-1}(a+p\beta)e^{-(a+p\beta)/Y} \ll p^{-1}\sum_\beta r(\beta)(Y+p\beta)e^{-\pi\beta} \ll 1$ by (ii). In $T'(k)^2 = \sum_{\beta,\beta'\ge1} r(\beta)r(\beta')r(k-p\beta)r(k-p\beta')$ the terms $\beta \ne \beta'$ are shifted convolutions with the shift $p|\beta-\beta'|$ and are $\ll d(\beta)d(\beta')e^{-\pi\max(\beta,\beta')/2}$ by (i), again $O(1)$ in total. The diagonal $\beta = \beta'$ is

$$D_Q = \sum_{\beta\ge1} r(\beta)^2 \sum_{a\ge0} r(a)^2 \frac{e^{-(a+p\beta)/Y}}{a+p\beta},$$

and by (iii) and partial summation, $\sum_{a\ge1} r(a)^2 F(a+h) = \int_1^\infty (4X\log X + O(X))(-F'(X+h))\,\mathrm{d}X$ for $F(t) = e^{-t/Y}/t$, where $-F'(t) = e^{-t/Y}(1/(Yt) + 1/t^2) > 0$; the $O(X)$ part is $\ll e^{-h/Y} + E_1(h/Y) \ll e^{-h/2Y}$, and the main part is, after an integration by parts and $X = Yu$,

$$4\int_1^\infty (\log X + 1)\frac{e^{-(X+h)/Y}}{X+h}\,\mathrm{d}X = 4\log Y\, E_1(h/Y) + O(e^{-h/2Y}).$$

With $h = p\beta$, $\log Y = \log p + O(1)$ and $h/Y \ge \pi\beta$, this gives $D_Q = 4\log p\sum_\beta r(\beta)^2 E_1(4\pi p\beta/Q) + O(1)$, and $\frac{1}{4\pi}(D_p + D_{4p}) = b_3\log p + O(1)$.

Main cusps, arcs. The terms $j = 0$ with $\Psi_0(c) - e^{-c}$ in place of e^{-c} are treated identically: $0 \le \Psi_0(c) - e^{-c} \le e^{-\sqrt{3}c/2}$ gives the same bounds with Y replaced by $2Y/\sqrt{3}$, and in the diagonal $\int_0^\infty \Psi_0(u+v_0)\,\mathrm{d}u/(u+v_0) = \int_{-1/2}^{1/2} E_1(sv_0)\,\mathrm{d}x$ by Fubini, since $\int_0^\infty e^{-s(u+v_0)}\mathrm{d}u/(u+v_0) = E_1(sv_0)$. Subtracting $E_1(v_0)$ gives b_4.

Main cusps, correlations. For $j \ge 1$ the term is $\frac{2}{Q}\sum_k a(k)a(k+jQ)\Psi_j(c)/c$ with $c = 2\pi(2k+jQ)/Q$ and $|\Psi_j(c)| \le e^{-\sqrt{3}c/2}$. Expanding both factors, every product containing M or two different arguments of r is $O(e^{-\sqrt{3}\pi j})$ by (i), (ii), with geometric decay in β, β'. The squares $r(k-p\beta)^2$ occur exactly when $\beta' - \beta = jQ/p$: $\beta' = \beta + j$ for $Q = p$ and $\beta' = \beta + 4j$ for $Q = 4p$, now with $\beta \ge 0$, the term $\beta = 0$ being the free part $r(k)$ paired with the term $\beta' = jQ/p$ of the tail. With $k = a + p\beta$, $a = pu$, partial summation with (iii) gives $\sum_a r(a)^2\Psi_j(c)/c = 4p\log p\int_0^\infty \Psi_j(c(u))\,\mathrm{d}u/c(u) + O(p)$, where $c(u) = 2\pi(2u+2\beta+j)$ for $Q = p$ and $c(u) = \pi(u+\beta+2j)$ for $Q = 4p$; the integrals are $\frac{1}{2}I_j(2\beta+j;2\pi)$ and $I_j(\beta+2j;\pi)$, and with the factors $2/Q$ the two widths give $4\log p\sum_\beta r(\beta)r(\beta+j)I_j(2\beta+j;2\pi)$ and $2\log p\sum_\beta r(\beta)r(\beta+4j)I_j(\beta+2j;\pi)$: the first two sums of b_6.

The cusp of denominator 2. Its width is $4p$, $w_d = p$ and $b = \frac{1}{p}(T^{\mathrm{odd}} - M^{\mathrm{odd}})$ (Part II, Lemma 6.12), where $T^{\mathrm{odd}}(k) = \sum_{\beta\ge1} r^{\mathrm{o}}(\beta)r^{\mathrm{o}}(k-p\beta)$ with $r^{\mathrm{o}}(a) = r(a)$ for $a \equiv 2 \pmod 4$ and 0 otherwise (odd x, y force $x^2 + y^2 \equiv 2 \pmod 8$, and conversely). The class is $\frac{1}{\pi}\sum_k (T^{\mathrm{odd}} -$

$M^{\mathrm{odd}})^2(k)e^{-\pi k/p}/k$ (height 1) and the same with $\Psi_0 - e^{-c}$; there is no free part, the M^{odd} terms and the off-diagonal terms are $O(1)$ as above, and the diagonal gives, by the second part of (iii) (coefficient 1 in place of 4) and $Y = p/\pi$, $\frac{1}{\pi}\log p \sum_{\beta\equiv 2(4)} r(\beta)^2 \int E_1(\pi\beta s)\mathrm{d}x + O(1) = b_5 \log p + O(1)$, height 1 and arcs together. Its correlations are the third sum of b_6, by the computation for $Q = 4p$.

The narrow cusps. At the widths 4 and 1 the weights are $e^{-\pi k}$ and $e^{-4\pi k}$; the coefficients are $a(k) = r(k) - M(k)$ for $k < p$ with $M(k) \ll k\sigma_{-1}(k)/p$, so these classes are $\frac{1}{4\pi}\sum_k r(k)^2 e^{-4\pi k/Q}/k + O(1/p)$, and the cusp of denominator $2p$ has $T^{\mathrm{odd}}(1, p; k) = 0$ for $k < 2(p+1)$ (Part II, Lemma 6.12), hence is $O(1/p)$ as well; the same holds for their arc and correlation terms.

Adding b_1 (Lemma 6.3 of Part II, as in Section 2.1) and b_2 (Proposition 2.2) gives the theorem. The script `closed_form_B.py` evaluates $b_1, \dots, b_6$ (E_1 from `scipy`, the x- and v-integrals by quadrature, $\beta \le 40$, $j \le 8$) and compares them with the local slopes of the exact anatomy over $p = 2\cdot 10^5$, $4\cdot 10^5$: 0.05687/0.05676 (C_2), 0.00133/0.00134 (cusp of denominator 2 at height 1), 0.04073/0.04073 (arcs, main cusps), 0.00057/0.00058 (arcs, other cusps), $-0.00046/-0.00046$ (correlations) and 0.48204/0.48188 for the total; and the part of C_2 that remains after the diagonal is removed is -1.1691 ± 0.0001 at height 1 and -0.05762 through the arcs for all $10^4 \le p \le 4\cdot 10^5$. □

Remark 2.5. (a) The Siegel main term does not enter B: the sums $\sum r(k)\beta(k)$ and $\sum \beta(k)^2$ grow linearly, without a logarithm, which is Lemma 2.3(ii) and sharpens Part II, Lemma 6.4, by the factor $\log 2Y$ there. The whole of $B - b_1 - b_2 = 0.0702$ comes from the squares $r(k - p\beta)^2$, $\beta \ge 1$, i.e. from the representations $k = \alpha + p\beta$ of the coefficients $k \ge p$, through the exponential integral $E_1(4\pi p\beta/Q)$ of the position $p\beta/Y$ of the shifted copy of $\sum r(a)^2$ in the weight $e^{-k/Y}$; the terms $\beta = 1, 2$ carry all but 10^{-6} of it. (b) The same mechanism applies to the odd Heegner discriminants of Section 3, where $\sum_{a\le X} r_D(a)^2 = \frac{24}{D+1}X\log X + O(X)$ by Theorem 3.5: at the two wide cusps (widths Dp and p, so $4\pi p\beta/Q \in \{4\pi\beta/D, 4\pi\beta\}$) the tail adds $\frac{6}{\pi(D+1)}\sum_{\beta\ge 1} r_D(\beta)^2\big(E_1(4\pi\beta/D) + E_1(4\pi\beta)\big)\log p$ to $b_D \log p$, which for $D = 4$ is b_3. This needs the analogue of Lemma 2.3 for r_D, which is the same lattice-point argument for the form N_D; we have not written it out.

3. The theta series of the odd Heegner discriminants

3.1. **Setting.** Let D be one of the seven odd Heegner discriminants, $K = \mathbb{Q}(\sqrt{-D})$, $\chi = \chi_{-D}$ the Kronecker symbol, w_D the number of units ($w_3 = 6$, $w_D = 2$ otherwise), $N_D(x, y) = x^2 + xy + \frac{D+1}{4}y^2$ (Gram determinant D), $r_D(n) = \#\{N_D = n\} = w_D\sum_{d|n}\chi(d)$ for $n \ge 1$ (class number one), and $\theta_D(z) = \sum_{n\ge 0} r_D(n)e(nz) \in M_1(\Gamma_0(D), \chi)$. The Eisenstein series $E(z, s) = \sum_{\Gamma_\infty\backslash\Gamma_0(D)} \chi(d_\gamma)j(\gamma, z)^{-1}\operatorname{Im}(\gamma z)^s$ has at $s = 0$ the expansion $1 + \frac{2}{L(0,\chi)}\sum \sigma_\chi(n)e(nz)$ with $L(0, \chi) = 2h/w_D = 2/w_D$ (Hecke), so $E(z, 0) = \theta_D$ without any dimension argument. For a prime $p \ne D$, $\Theta_{1,p} = \theta_D(z)\theta_D(pz) = E + G \in M_2(\Gamma_0(Dp))$, $T(k) = \sum_{\alpha+p\beta=k} r_D(\alpha)r_D(\beta)$, $a_G(k) = T(k) - M(k)$. The level Dp is square-free: $\Gamma_0(Dp)$ has the four cusps $W_Q\infty$, $Q \in \{1, D, p, Dp\}$, of widths $Dp, p, D, 1$, and no cusp of denominator 2δ. We write $|_k$ for the weight-k slash operator normalised by $(\det\gamma)^{k/2}$, so that scalars act trivially and $f(mz) = m^{-k/2}(f|_k B_m)(z)$, $B_m = \left(\begin{smallmatrix} m & 0 \\ 0 & 1 \end{smallmatrix}\right)$.

3.2. **The Siegel main term.**

Theorem 3.1. *$M(k) = \dfrac{4\pi^2 k}{Dp}\prod_\ell \beta_\ell(k)$, where $\beta_\ell(k) = (1-\ell^{-2})\sigma_{-1}(\ell^{v_\ell(k)})$ for every prime $\ell \nmid Dp$, including $\ell = 2$, and, with $v = v_\ell(k)$,*

$$\beta_p(k) = (1+\chi(p)) - \chi(p)\Big(1+\frac{1}{p}\Big)p^{-v}, \qquad \beta_D(k) = (1+\varepsilon) - \varepsilon\Big(1+\frac{1}{D}\Big)D^{-v},$$

where $\varepsilon = \left(\frac{-p}{D}\right) = -\chi_{-D}(p)$.

Proof. The quaternary form $Q = N_D(x,y)+pN_D(z,w)$ has Gram determinant D^2p^2, and Siegel's theorem [10] gives $r(\operatorname{gen} Q, k) = 4\pi^2 k\prod_\ell \beta_\ell(k)/\sqrt{D^2p^2}$ with $\beta_\ell(k) = \lim_a \ell^{-3a}\#\{x \bmod \ell^a : Q(x) \equiv k\}$; the genus theta series is the Eisenstein part of $\Theta_{1,p}$. For $\ell \nmid Dp$ the form is unimodular over $\mathbb{Z}_\ell$ with square discriminant, which gives the standard density; this includes $\ell = 2$ because N_D is even unimodular over $\mathbb{Z}_2$, $pN_D \cong N_D$ there (p a unit; the Arf invariant is insensitive to unit scaling), and $N_D \oplus N_D$ has Arf invariant 0, i.e. is the sum of two hyperbolic planes, whatever the class of N_D modulo 2.

At $\ell = p$: the solutions with $(x,y) \not\equiv (0,0) \pmod p$ are non-singular in (x,y), so each solution modulo p lifts to p^{a-1} solutions modulo p^a; modulo p the equation $N_D(x,y) \equiv t$ has $p - \chi(p)$ solutions for $p \nmid t$ and $1+(p-1)(1+\chi(p))$ for $p \mid t$. With (z,w) free this gives $\beta_p(k) = 1-\chi(p)/p$ for $p \nmid k$ and the contribution $(1+\chi(p))(1-1/p)$ for $p \mid k$. The solutions with $x \equiv y \equiv 0 \pmod p$, $x = px'$, $y = py'$, satisfy $pN_D(x',y') + N_D(z,w) \equiv k/p \pmod{p^{a-1}}$, the same form with the blocks interchanged; counting (x',y') modulo p^{a-1} and the p^2 lifts of (z,w) gives $\beta_p(k/p)/p$. Hence $\beta_p(k) = (1+\chi(p))(1-1/p) + \beta_p(k/p)/p$ for $p \mid k$, which iterates to the closed form.

At $\ell = D$: since 2 is invertible, $N_D(x,y) = (x+y/2)^2 + \frac{D}{4}y^2 \cong u^2 + Dv^2$ over $\mathbb{Z}_D$, so $Q \cong u^2 + ps^2 + D(v^2+pt^2)$. The solutions with $(u,s) \not\equiv 0 \pmod D$ are non-singular in (u,s); modulo D the binary form u^2+ps^2, of discriminant $-4p$, represents $t \not\equiv 0$ exactly $D-\varepsilon$ times and $t \equiv 0$ exactly $1+(D-1)(1+\varepsilon)$ times, $\varepsilon = (-p/D)$. The solutions with $u \equiv s \equiv 0$ reduce to $D(u'^2+ps'^2) + (v^2+pt^2) \equiv k/D \pmod{D^{a-1}}$, the same form, with the factor $D^2/D^3 = 1/D$. Hence $\beta_D(k) = 1-\varepsilon/D$ for $D \nmid k$ and $\beta_D(k) = (1+\varepsilon)(1-1/D)+\beta_D(k/D)/D$ for $D \mid k$. Finally $(-p/D) = (-1/D)(p/D) = -(p/D) = -(-D/p) = -\chi_{-D}(p)$ by quadratic reciprocity, $D \equiv 3 \pmod 4$. □

3.3. **Atkin–Lehner signs and the spectral support.**

Theorem 3.2. *$\Theta_{1,p}|_2 W_p = \chi(p)\Theta_{1,p}$, $\Theta_{1,p}|_2 W_D = -\chi(p)\Theta_{1,p}$ and $\Theta_{1,p}|_2 W_{Dp} = -\Theta_{1,p}$, for any choice of the Atkin–Lehner matrices [1] of level Dp.*

Proof. Write $W_p = \left(\begin{smallmatrix} pa & b \\ Dpc & pd\end{smallmatrix}\right)$, $pad - Dbc = 1$. Then $W_p = \gamma B_p$ with $\gamma = \left(\begin{smallmatrix} a & b \\ Dc & pd\end{smallmatrix}\right) \in \Gamma_0(D)$, so $\theta_D|_1 W_p = \chi(pd)p^{1/2}\theta_D(pz)$; and $B_pW_p = p\left(\begin{smallmatrix} pa & b \\ Dc & d\end{smallmatrix}\right)$ with the second factor in $\Gamma_0(D)$, so $\theta_D(p\,\cdot)|_1 W_p = p^{-1/2}\chi(d)\theta_D$. The product is $\chi(p)\Theta_{1,p}$.

For W_D: the dual of the lattice $(\mathcal{O}_K, N_D)$ is $\mathcal{O}_K/\sqrt{-D}$, so Poisson summation gives $\theta_D(-1/z) = D^{-1/2}(-iz)\theta_D(z/D)$ and $\theta_D(-1/(Dz)) = -i\sqrt{D}z\,\theta_D(z)$, i.e. $\theta_D|_1 W_0 = -i\theta_D$ for $W_0 = \left(\begin{smallmatrix} 0 & -1 \\ D & 0\end{smallmatrix}\right)$. Every matrix $W = \left(\begin{smallmatrix} Da' & b' \\ Dc' & Dd'\end{smallmatrix}\right)$ of determinant D is $W_0\gamma'$ with $\gamma' = \left(\begin{smallmatrix} c' & d' \\ -Da' & -b'\end{smallmatrix}\right) \in \Gamma_0(D)$, whence $\theta_D|_1 W = -i\chi(-b')\theta_D$. Now $W_D = \left(\begin{smallmatrix} Da & b \\ Dpc & Dd\end{smallmatrix}\right)$, $Dad - pbc = 1$, is such a matrix with $b' = b$, and $B_pW_D = \left(\begin{smallmatrix} Da & pb \\ Dc & Dd\end{smallmatrix}\right)B_p$ exhibits $\theta_D(p\,\cdot)|_1 W_D = p^{-1/2}(\theta_D|_1 W')|_1 B_p$ with W' such a matrix with $b' = pb$. Therefore $\Theta_{1,p}|_2 W_D = (-i)^2\chi(-b)\chi(-pb)\Theta_{1,p} = -\chi(p)\Theta_{1,p}$, using $(b,D) = 1$; and $W_{Dp} \equiv W_DW_p$ modulo $\Gamma_0(Dp)$ gives -1. □

Theorem 3.3 (Spectral support)**.** *Let φ be a newform of weight 2 and level Dp. If $\langle \Theta_{1,p}, \varphi\rangle \neq 0$ then $\eta_p(\varphi) = \chi(p)$, $\eta_D(\varphi) = -\chi(p)$, i.e. $a_\varphi(D) = \chi_{-D}(p)$, $a_\varphi(p) = -\chi_{-D}(p)$, and the root*

number of φ is $+1$. For a newform f of level p and $d \mid D$, $\langle \Theta_{1,p}, f(d\cdot)\rangle \neq 0$ requires $\eta_p(f) = \chi(p)$, i.e. $a_f(p) = -\chi_{-D}(p)$. For a newform f of level D and $d \mid p$, $\langle \Theta_{1,p}, f(d\cdot)\rangle = C_D(1+\chi(p))L(1,f)L(1,f\otimes\chi)/d$ vanishes when $\chi_{-D}(p) = -1$.

Proof. $\langle \Theta, \varphi\rangle = \langle \Theta|W, \varphi|W\rangle$ for $W = W_p, W_D$ and Theorem 3.2 give $\chi(p)\eta_p(\varphi) = 1$ and $-\chi(p)\eta_D(\varphi) = 1$; $a_\varphi(\ell) = -\eta_\ell(\varphi)$ for $\ell \,\|\, Dp$, and the root number is $-\eta_D\eta_p$. For f of level p, Part II, Lemma 6.8 (with $Q = p$) gives $f(d\cdot)|_2 W_p = \eta_p(f) f(d\cdot)$, and the same argument applies; W_D exchanges $f(z)$ and $f(Dz)$ up to a factor and gives no further condition. For f of level D the formula is the oldform computation in the proof of Part II, Theorem 6.7(ii), with D in the role of 4 and p in the role of n: the brackets there give $\sum_{Q_2|d}\chi(Q_2)\prod_{\ell|p/d}(1+\chi(\ell)) = 1+\chi(p)$ in both cases $d = 1, p$. □

In contrast with $D = 4$, where every newform of level $4n$ has $\eta_4 = -1$ (Part II, Proposition 6.17), so that the newform support is empty when $\chi_{-4}(n) = -1$, both eigenvalues of W_D occur among the newforms of level Dp, the local component at D being the Steinberg representation or its unramified quadratic twist: there is no inert vanishing. The support subspace $\{\eta_D = -\chi(p),\ \eta_p = \chi(p)\}$ has dimension $\frac{1}{4}(\dim S^{\mathrm{new}} + \chi(p)\operatorname{tr} W_p - \chi(p)\operatorname{tr} W_D - \operatorname{tr} W_{Dp})$, the traces taken on $S_2^{\mathrm{new}}(\Gamma_0(Dp))$; these traces are sums of class numbers of size $O(\sqrt{Dp}\log Dp)$ by the trace formulae for the Atkin–Lehner involutions (Yamauchi [12] for the Fricke involution W_{Dp}; Martin [9] for the dimensions of the eigenspaces with prescribed signs at square-free level), so the support is asymptotically a quarter of the new subspace, i.e. half of the forms with root number $+1$, cut down further by $L(1,\varphi)L(1,\varphi\otimes\chi) \neq 0$. The oldforms from level D exist for $D \in \{11, 19, 43, 67, 163\}$ (genus $1, 1, 3, 5, 13$) and not for $D = 3, 7$; at the level 22 ($\chi_{-11}(2) = -1$) the cuspidal part vanishes identically, $\theta_{11}(z)\theta_{11}(2z) = E$, while at the level 55 ($\chi_{-11}(5) = +1$) the component of f_{11} is $f_{11}(z) + 5f_{11}(5z)$ up to a scalar and carries 25% of the norm.

3.4. **The Rankin–Selberg formula.**

Theorem 3.4. *For a newform φ of weight 2 and level Dp,*

$$\langle \Theta_{1,p}, \varphi\rangle = C_D\,\chi(p)\,\eta_p(\varphi)\,L(1,\varphi)\,L(1,\varphi\otimes\chi_{-D}), \qquad C_D = \frac{w_D}{4\pi L(1,\chi_{-D})} = \frac{w_D^2\sqrt{D}}{8\pi^2};$$

$C_3 = \frac{9\sqrt{3}}{2\pi^2} = 0.7897$, $C_D = \frac{\sqrt{D}}{2\pi^2}$ *for $D \geq 7$, against $C_4 = \frac{4}{\pi^2}$ and $C_8 = \frac{\sqrt{2}}{\pi^2}$.*

Proof. The proof of Part II, Theorem 6.7(i) carries over: $\theta_D = E(z,0)$; the cusps of $\Gamma_0(Dp)$ over ∞ of $\Gamma_0(D)$ are $W_Q\infty$, $Q \mid p$; unfolding gives $I(s) = \sum_{Q|p}\chi(Q)\Gamma(s+1)(4\pi)^{-s-1}\sum_j r_D(j)c_Q((p/Q)j)((p/Q)j)^{-s-1}$, with $\chi(a)\chi(d) = \chi(Q)$ from $Qad \equiv 1 \pmod D$ and $c_Q(k) = \eta_Q(\varphi)a_\varphi(k)$; the Dirichlet series is $w_D L(w,\varphi)L(w,\varphi\otimes\chi)/(L(2w-1,\chi)\prod_{\ell|p}(1-\chi(\ell)\ell^{1-2w}))$, the local factor at $\ell = D$ being $(1-a_\varphi(D)x)^{-1} = L_D(w,\varphi)L_D(w,\varphi\otimes\chi)$ because $\chi(D) = 0$ and $\sigma_\chi(D^j) = 1$. At $s = 0$ the constant is $w_D/(4\pi L(1,\chi))$, and $\sum_{Q|p}\chi(Q)\eta_Q(\varphi)a_\varphi(p/Q)/(p/Q) = \chi(p)\eta_p(\varphi)(1-\chi(p)/p)$ cancels the local factor. The class number formula $L(1,\chi_{-D}) = 2\pi h/(w_D\sqrt{D})$ with $h = 1$ gives the second expression. □

3.5. **The norm.**

Theorem 3.5. *Let $D_D(s) = \sum_k r_D(k)^2 k^{-s}$. Then $D_D(s) = w_D^2\,\zeta(s)^2 L(s,\chi)^2/\big((1+D^{-s})\zeta(2s)\big)$ and*

$$S_D(Y) := \sum_{k\geq 1}\frac{r_D(k)^2}{k}e^{-k/Y} = \frac{12}{D+1}(\log Y)^2 + c_1^{(D)}\log Y + c_2^{(D)} + O(Y^{-1/2}),$$

$$c_1^{(D)} = \frac{24}{D+1}\Big(\gamma + 2\frac{L'}{L}(1,\chi) - 2\frac{\zeta'}{\zeta}(2) + \frac{\log D}{D+1}\Big),$$

$$\frac{L'}{L}(1,\chi_{-D}) = \gamma + \log 2 + \log\pi - \frac{w_D}{2}\sum_{a=1}^{D-1}\chi(a)\log\Gamma\Big(\frac{a}{D}\Big).$$

Consequently the regions above height 1 *at the two wide cusps give*

$$\|G_{1,p}^{(D)}\|^2 \;\ge\; \frac{1}{4\pi}\Big(S_D\Big(\frac{Dp}{4\pi}\Big)+S_D\Big(\frac{p}{4\pi}\Big)\Big) - O(\log p) = \frac{6}{\pi(D+1)}(\log p)^2 + b_D\log p + O(1),$$

$$b_D = \frac{6}{\pi(D+1)}\log\frac{D}{16\pi^2} + \frac{c_1^{(D)}}{2\pi},$$

with $b_3 = 0.713$, $b_7 = 0.207$, $b_{11} = 0.133$, $b_{19} = 0.121$, $b_{43} = 0.143$, $b_{67} = 0.150$, $b_{163} = 0.142$.

Proof. $r_D/w_D = \sigma_\chi$ is multiplicative with $\sigma_\chi(\ell^j) = j+1$, $\frac12(1+(-1)^j)$, 1 for $\chi(\ell) = 1, -1, 0$, so the Euler factor of $\sum\sigma_\chi(k)^2k^{-s}$ at $\ell \ne D$ is that of $\zeta^2L^2/\zeta(2s)$, and at $\ell = D$ it is $(1-x)^{-1}$ against $(1+x)/(1-x)$, whence the factor $(1+D^{-s})^{-1}$. At $s=1$: $w_D^2L(1,\chi)^2\cdot\frac{D}{D+1}\cdot\frac{6}{\pi^2} = \frac{24}{D+1}$ by the class number formula, and the residue computation of Part II, Lemma 6.3 gives S_D. The value of $\frac{L'}{L}(1,\chi)$ follows from the functional equation $\Lambda(s) = (D/\pi)^{s/2}\Gamma(\frac{s+1}{2})L(s,\chi) = \Lambda(1-s)$ and $L(s,\chi) = D^{-s}\sum_a\chi(a)\zeta(s,a/D)$ with $\zeta'(0,a) = \log\Gamma(a) - \frac12\log 2\pi$: $\frac{L'}{L}(1) = -\log\frac{D}{\pi} - \frac12\psi(\frac12) - \frac12\psi(1) - \frac{L'}{L}(0)$ and $\frac{L'}{L}(0) = -\log D + \frac{w_D}{2}\sum_a\chi(a)\log\Gamma(a/D)$, since $L(0,\chi) = 2/w_D$. The lower bound for the norm is Part II, Lemma 6.2 at the cusps $W_1\infty$ and $W_D\infty$ (widths Dp and p), which carry $\pm G_{1,p}$ by Theorem 3.2, with the correction by the main term and by the coefficients $k \ge p$ bounded as in Part II, Theorem 1.5(iii). □

3.6. **The even discriminants, and the volume.** For $D = 4$ ($\theta_4 = \theta^2$, $w_4 = 4$) and $D = 8$ ($\theta_8(z) = \sum e((x^2+2y^2)z)$, $w_8 = 2$) the theta series are again $E_1(z,\chi_{-D})$, and the level Dp has, besides the Atkin–Lehner cusps, the cusps of denominator 2δ ($D = 4$) and $2\delta, 4\delta$ ($D = 8$) of Part II, Section 6.6. The two ingredients that fix the constants are the same, $r_D = w_D\sigma_\chi$ and $w_D^2L(1,\chi_{-D})^2 = 4\pi^2/D$; hence $C_D = w_D^2\sqrt{D}/(8\pi^2)$ for all nine discriminants ($C_8 = \sqrt2/\pi^2$ with the same proof, once the cusps of $\Gamma_0(8p)$ over ∞ of $\Gamma_0(8)$ are listed, which we have not written out), and the residue of D_D at $s=1$ is $\frac{4\pi^2}{D}\cdot\frac{6}{\pi^2}\prod_{\ell\mid D}(1+\ell^{-1})^{-1} = 24/\psi(D)$, the Euler factor at each ramified prime being $(1+\ell^{-s})^{-1}$. The two wide Atkin–Lehner cusps (widths Dp and p, carrying $\pm G_{1,p}$ by the computation of Theorem 3.2, using $\theta_D(-1/(Dz)) = -i\sqrt{D}z\,\theta_D(z)$ for $D = 8$ as well) therefore give, uniformly for the nine discriminants,

$$\|G_{1,p}^{(D)}\|^2 \;\ge\; \frac{6}{\pi\,\psi(D)}(\log p)^2 + O(\log p) = \frac{2}{\mathrm{vol}(\Gamma_0(D)\backslash\mathbb{H})}(\log p)^2 + O(\log p),$$

since $\mathrm{vol}(\Gamma_0(D)\backslash\mathbb{H}) = \frac{\pi}{3}\psi(D)$: $\frac{1}{\pi}$ for $D = 4$ (Part II, Theorem 6.14, where it is the asymptotic), $\frac{1}{2\pi}$ for $D = 8$, $\frac{6}{\pi(D+1)}$ for the odd D. The growth constant is the reciprocal volume of the modular curve of the *discriminant*, not of the level: in the Rankin–Selberg normalisation $\sum_{k\le Y}|a(k)|^2/k \sim 16\pi^2\|G\|^2Y/\mathrm{vol}(\Gamma_0(Dp)\backslash\mathbb{H})$, the index $[\Gamma_0(D):\Gamma_0(Dp)] = p+1$ cancels against the sparseness $1/p$ of the sequence $p\beta$. The value $\sqrt{D}/(2\pi)$ that one might guess from the discriminant alone is wrong already at $D = 3$ (0.276 against $6/(4\pi) = 0.4775$, fitted 0.4815). The proof of the asymptotic formula, Part II, Theorem 6.14, carries over to the odd D with four Atkin–Lehner classes and no odd class; we have not written it out, and state (v) as a lower bound.

4. Numerical verification

The script `heegner_theta_unified.py` (self-contained: Kronecker symbol, r_D, T, M, counts of local densities, Atkin–Lehner check at a base point, Hecke decomposition by T_2 then

T_3, T_5, T_7, T_{13} where eigenvalues coincide, central values by the smoothed functional equation, norms by Rankin–Selberg and by the cusp sums) was run for each odd D. The checks, uniformly: the closed-form densities agree with direct counts modulo D^a and p^a ($p \le 13$) to $2 \cdot 10^{-16}$; at the level 6, $T = M$ for all $k \le 2^{16}$ (no cusp forms); at the level 14, $T - M = \frac{2}{3} f_{14}$, at the level 15, $T - M = \frac{9}{2} f_{15}$, at the level 22, $T - M = 0$; the three Atkin–Lehner signs hold to 10^{-15} for $p \le 13$.

D	w_D	$L(1,\chi)$	$\frac{L'}{L}(1,\chi)$	C_D	$\frac{12}{D+1}$	$c_1^{(D)}$	$c_2^{(D)}$	$\frac{6}{\pi(D+1)}$	fit a	newforms, RS ratios
3	6	0.604600	0.368282	0.789720	3	16.370	13.059	0.4775	0.4815	21: [0.9944, 1.0048]
7	2	1.187410	0.015636	0.134035	1.5	5.975	1.450	0.2387	0.2408	14: [0.9966, 1.0047]
11	2	0.947226	−0.084218	0.168022	1	3.497	0.492	0.1592	0.1601	20: [0.9952, 1.0029]
19	2	0.720731	−0.084914	0.220824	0.6	2.033		0.0955	0.0961	23: [0.9965, 1.0047]
43	2	0.479088	0.249482	0.332204	0.2727	1.255		0.0434	0.0443	13: [0.9936, 1.0038]
67	2	0.383746	0.658364	0.414695	0.1765	1.093		0.0281	0.0266	(not resolved)
163	2	0.246050	2.168327	0.646797	0.0732	0.891		0.0116	0.0128	(not resolved)

TABLE 2. The constants of Theorems 3.4 and 3.5, the fitted coefficient of $(\log p)^2$ in $\Lambda_1(G_{1,p}^{(D)})$ over $p \in \{101, 1009, 10007, 100003\}$, and the newform components of $G_{1,p}^{(D)}$ resolved by the Hecke decomposition for $p \le 37$ ($D = 3$) and $p \le 19$ (otherwise), all of which obey Theorem 3.3, with the range of the ratios of the two sides of Theorem 3.4 (the twist has level $D^2 p$ and root number $+1$ in every case). $L(1,\chi)$ and $\frac{L'}{L}(1,\chi)$ agree with mpmath for $D \le 19$. For $D = 67, 163$ the dimension of $S_2(\Gamma_0(Dp))$ ($\ge 33, \ge 81$) is beyond the Prony recurrence of order ≤ 11, and the support is not tested. Files `results/heegner_theta_unified_*.txt`; floating point.

	$D=3$		$D=7$		$D=11$		$D=19$		$D=43$		$D=67$		$D=163$	
p	Λ_1	Λ_h	Λ_1	Λ_h	Λ_1	Λ_h	Λ_1	Λ_h	Λ_1	Λ_h	Λ_1	Λ_h	Λ_1	Λ_h
101	11.62	12.51	5.29	5.77	3.97	4.36	3.29	3.64	3.08	3.34	3.01	3.21	2.88	2.98
1009	25.97	27.30	12.26	13.01	8.78	9.40	6.50	7.02	4.98	5.34	4.53	4.80	3.76	3.88
10007	45.39	47.15	21.77	22.78	15.28	16.12	10.73	11.43	7.35	7.80	6.30	6.63	4.78	4.92
100003	69.92	72.12	33.83	35.12	23.48	24.54	15.98	16.84	10.19	10.73	8.39	8.78	5.93	6.10

TABLE 3. The sandwich $\Lambda_1 \le \|G_{1,p}^{(D)}\|^2 \le \Lambda_h$, $h = \sqrt{3}/2$, over the four cusps of $\Gamma_0(Dp)$ (Part II, Lemma 6.11), with $k \le 12p$. The Rankin–Selberg estimates of the norm with the window 2^{21} are 11.89, 5.44, 4.09, 3.40, 3.24, 3.34, 3.49 at $p = 101$ and lie inside the sandwich there; at $p = 1009$ they have not converged for $D \ge 19$ ($X/L < 110$). Floating point.

The two curiosities of the data. At the level 77 the same three newforms carry $G_{1,11}^{(7)}$ and $G_{1,7}^{(11)}$ with shares $(0.001, 0.019, 0.981)$ and $(0.502, 0.495, 0.005)$ (Figure 3): the support conditions of Theorem 3.3 coincide for the two pairs (D, p), and the two twisted central values distribute the norm in opposite ways. And at $D = 163$ the term $b_D \log p$ with $b_{163} \approx 0.24$ (fitted) exceeds the leading term $0.0116(\log p)^2$ for $p < 10^9$: the suppression of the norm by the factor 41 is, in the range of any computation, a suppression of $(\log p)^2$ only.

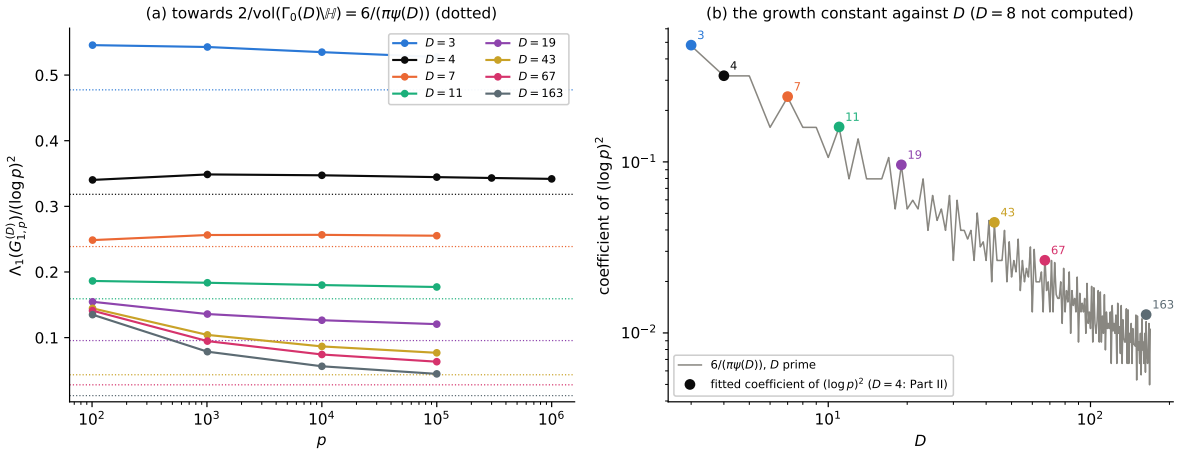


FIGURE 2. (a) $\Lambda_1(G_{1,p}^{(D)})/(\log p)^2$ against p for $D = 4$ (Part II, Section 6.6(f)) and the seven odd discriminants, with the limits $2/\operatorname{vol}(\Gamma_0(D)\backslash\mathbb{H}) = 6/(\pi\psi(D))$ dotted; the approach is slow, $\frac{6}{\pi\psi(D)} + b_D/\log p$, with b_D of Theorem 3.5 plus the corrections. (b) The fitted coefficient of $(\log p)^2$ against D, with the curve $6/(\pi\psi(D))$ through the primes; a factor 41 between $D = 3$ and $D = 163$. $D = 8$ has not been computed. Script `plot_heegner_volume_scaling.py`; floating point.

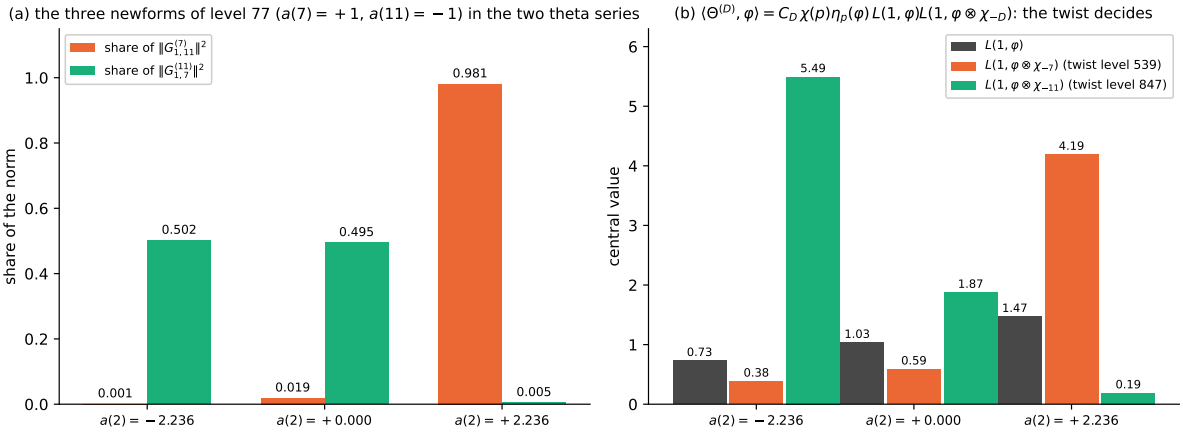


FIGURE 3. The level 77 is $7 \cdot 11$ in two ways. The three newforms of level 77 with $a(7) = +1$, $a(11) = -1$ (and $a(2) = -\sqrt{5}, 0, \sqrt{5}$) are exactly the support of both $G_{1,11}^{(7)}$ ($\chi_{-7}(11) = +1$: $a(7) = \chi_{-7}(11)$, $a(11) = -\chi_{-7}(11)$) and $G_{1,7}^{(11)}$ ($\chi_{-11}(7) = -1$: $a(11) = \chi_{-11}(7)$, $a(7) = -\chi_{-11}(7)$). (a) Their shares of the two norms, $0.001, 0.019, 0.981$ against $0.502, 0.495, 0.005$. (b) The central values that decide them through Theorem 3.4: the same $L(1, \varphi)$, but the twists by χ_{-7} (level 539) and by χ_{-11} (level 847) favour opposite forms. Script `plot_level77_duality.py`; floating point.

5. Sieve limits and the Eisenstein dispersion form

5.1. **Does a large discriminant help the sieve?** For the sequence $r_D(n)r_D(N-n)$, the norm $\|G^{(D)}_{m,m'}\|^2 \sim \frac{6}{\pi\psi(D)}(\log mm')^2$ enters the remainder through the uniform bound $|a_G(N)| \ll N^{1/2+\varepsilon}\|G\|$ (Part I, Theorem 2.3, whose proof is independent of D), and it decreases like $1/D$. This does not help. The main term decreases as well: the mean value of r_D is $w_D L(1,\chi_{-D}) = 2\pi/\sqrt{D}$, so $X_N = \sum_n r_D(n)r_D(N-n) \asymp \frac{4\pi^2}{D}N\mathfrak{S}_D(N)$ is of size N/D, while the remainder for one pair (d,e) of sieve moduli is $\ll N^{1/2+\varepsilon}(\log de)/\sqrt{\psi(D)}$; the ratio remainder/main is $\asymp \sqrt{D}\,N^{-1/2}\log de$ for prime D, larger by $\sqrt{D}$, not smaller. And the level of distribution is an exponent while D is fixed: the proof of the level $1/2$ (Part I, Theorem 3.2) loses only constants and N^{ε} in D, so the level is $1/2$ for every D, and the sieve dimension is 2 for every D, since $\sum_{p\le z}\omega_D(p)/p = 2\log\log z + O(1)$ with $\omega_D(p) = 2(1+\chi_{-D}(p))$ on average. What a large discriminant changes is the constant in the final count, $\asymp N/(D(\log N)^2)$, and, for the odd D, the shape of the fundamental domain, which simplifies the norm and has no bearing on the dispersion problem.

5.2. **The circle modulo c for the Eisenstein integers.**

Lemma 5.1. *Let $(c,3)=1$, c odd or even, $(a,c)=1$, $h\in\mathbb{Z}^2$, $|h|_3^2 = h_1^2 - h_1h_2 + h_2^2$, c'^* the product of the primes dividing c' to an odd power, and $\chi=\chi_{-3}$ ($\chi(2)=-1$). Then*

$$F_c(a;h) := \sum_{\substack{x,y \bmod c\\ N_3(x,y)\equiv a\ (c)}} e\Big(\frac{h_1x+h_2y}{c}\Big) = \sum_{g\mid(c,h_1,h_2)} g\,\chi\Big(\Big(\frac{c}{g}\Big)^*\Big) S\Big(a,\ \overline{3}\Big|\frac{h}{g}\Big|_3^2;\ \frac{c}{g}\Big).$$

Proof. As in Part III, Lemma 3.1. Detect the congruence by $\frac{1}{c}\sum_{t \bmod c} e(t(N_3(x,y)-a)/c)$; for $(t,c)=g$ the sum over (x,y) vanishes unless $g\mid h$, and for $(t,c)=1$ one completes the square: with the Gram matrix $A=\left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$ the substitution $x\mapsto x-\bar tA^{-1}h$ (legitimate modulo c, $\det A=3$ being invertible also at 2) gives $\sum_{x,y}e((tN_3+h\cdot x)/c) = e(-\bar t h^{\mathsf T}A^{-1}h/(2c))G_A(t;c)$ with $h^{\mathsf T}A^{-1}h/2 = |h|_3^2/3$; for odd c, $G_A(t;c) = G(t;c)G(3\bar 4 t;c) = c\,\chi(c^*)$ after diagonalising $N_3 = (x+y/2)^2 + \frac34 y^2$, the twisting Jacobi symbols of the Chinese remainder theorem squaring to 1; at the prime 2 the form is unimodular and $G_A(t;2^\nu) = (-1)^\nu 2^\nu = \chi((2^\nu)^*)2^\nu$. The script `eisenstein_circle_check.py` verifies the identity for all a, h and all $c\le 40$ coprime to 3, odd and even (285 902 cases). □

5.3. **The dispersion form.** Let m,m' be coprime square-free integers prime to $3N$. Since the ideal $(\sqrt{-3})$ is principal of norm 3, $r_3(3\beta') = r_3(\beta')$, and the terms of $T(m,m';N)$ with $3\mid\beta$ are $T(m,3m';N)$: with $T^\flat(m,m';N) = \sum_{3\nmid\beta} r_3(\alpha)r_3(\beta)$,

$$a_{G_{m,m'}}(N) - a_{G_{m,3m'}}(N) = T^\flat(m,m';N) - M^\flat(m,m';N),$$

the analogue of $a_G(N) - a_G(N/2) = T^{\mathrm{odd}} - M^{\mathrm{odd}}$ of Part III. In $T^\flat$, $r_3(\beta)\ne 0$ forces $\beta\equiv 1 \pmod 3$, so $\chi(\beta/k)=\chi(k)$ and $r_3(\beta) = 12\sum_{k\mid\beta,\,k<\sqrt\beta}\chi(k) + 6\chi(\sqrt\beta)[\beta=\square]$, the divisors k being prime to 3 and to m but possibly even. Hence

$$T^\flat(m,m';N) = 12\sum_{\substack{k<\sqrt{N/m'}\\ 3\nmid k}}\chi(k)\,\mathcal{N}_k + 6\sum_j \chi(j)\,r_3^\flat\Big(\frac{N-m'j^2}{m}\Big),$$

$$\mathcal{N}_k = \#\Big\{(x,y):\ N_3(x,y)\equiv N\bar m\ (m'k),\ mN_3(x,y)\not\equiv N\ (3),\ 0<N_3(x,y)<Y_k\Big\},$$

$Y_k = (N - m'k^2)/m$. The condition modulo 3 is the analogue of the parity condition $x + y$ odd of Part III: Poisson summation on the classes modulo $3c$, $c = m'k$, gives

$$\mathcal{N}_k = \frac{Y_k}{(3c)^2} \sum_{h \in \mathbb{Z}^2} \hat{w}\Big(\frac{h\sqrt{Y_k}}{3c}\Big)\, \epsilon_3(h)\, F_c(N\bar{m};\ \bar{3}h),$$

where $\bar{3}h$ is the Chinese-remainder twist and $\epsilon_3(h)$ the Fourier transform of the admissible classes modulo 3: as $N_3(x,y) \equiv (x-y)^2 \pmod 3$, the condition is $x \equiv y$ if $N\bar{m} \equiv 1$, $x \not\equiv y$ if $N\bar{m} \equiv 0$, and empty if $N\bar{m} \equiv 2 \pmod 3$, so that $\epsilon_3(h) = 3[h_1 + h_2 \equiv 0\ (3)]$, $9[h \equiv 0\ (3)] - 3[h_1 + h_2 \equiv 0\ (3)]$, 9, respectively. Since $|h|_3^2 \equiv (h_1 + h_2)^2 \pmod 3$, the condition $h_1 + h_2 \equiv 0 \pmod 3$ is $3 \mid |h|_3^2$: exactly the condition $h_1 \equiv h_2 \pmod 2$, i.e. $2 \mid |h|^2$, of Part III, with the ramified prime of the field in the role of 2. By Lemma 5.1 the leading term $g = 1$ of $F_c(N\bar{m}; \bar{3}h)$ is $\chi((m'k)^*)S(N\bar{m}, \overline{27}|h|_3^2; m'k) = \chi((m'k)^*)S(\overline{27m}N, |h|_3^2; m'k)$, and $\chi(k)\chi((m'k)^*) = \chi(m')$ for square-free m'.

Proposition 5.2. *The leading term ($g = 1$) of $a_{G_{m,m'}}(N) - a_{G_{m,3m'}}(N)$ is*

$$12\,\chi(m') \sum_{\substack{k < \sqrt{N/m'} \\ 3 \nmid k}} \frac{Y_k}{(3m'k)^2} \sum_{h \neq 0} \hat{w}\Big(\frac{h\sqrt{Y_k}}{3m'k}\Big)\, \epsilon_3(h)\, S\big(\overline{27m}\,N,\ |h|_3^2;\ m'k\big),$$

and the remainder sum of the sieve at the level N^θ has as its leading term the trilinear form $\sum_{m'} \Lambda'_{m'} \sum_{c \equiv 0\,(m')} \sum_m \Lambda_m \sum_j b_{m,j}(c) S(\overline{27m}N, j; c)$ over the moduli $c = m'k$ prime to 3, odd or even, with $j = |h|_3^2 \ll mm'$ a norm from $\mathbb{Z}[\zeta_3]$, whose m-th piece is, by the Chinese remainder theorem, the bilinear form in the two moduli

$$\mathfrak{B}_m^{(3)} = \sum_{m'} \Lambda'_{m'} \sum_{k \le K} \sum_h w_{k,m'}(h)\, S\big(\overline{27m}N\bar{k},\ |h|_3^2\bar{k};\ m'\big)\, S\big(\overline{27m}N\overline{m'},\ |h|_3^2\overline{m'};\ k\big).$$

Comparison with the Gaussian case. Nothing in the structure changes. (i) The fixed argument is the single integer N, carried by a unit ($\overline{27m}$ in place of $\overline{16m}$) modulo the modulus, so the obstacle of Part III, Section 3.4, the absence of an average over it and the inapplicability of the Kuznetsov formula, is the same. (ii) The sizes are the same: $|h| \ll m'k\sqrt{m/N}$, the h-box has side $H \le \sqrt{mm'}$, below $\sqrt{q}$ for every modulus $q = [c_1, c_2]$ of a Poisson step, and the Cauchy–Schwarz arrangements (A1), (A2) of Part III, Section 4 give the same diagonals ($E^{1/4}N^{3/4}$ and $\sqrt{EN}$); there is no tighter grouping, because the groupings depend only on the lengths of the variables and on the Weil mean square of Kloosterman sums over the dual lattice, which is the same for the hexagonal lattice. (iii) The moduli $m'k$ range over all k prime to 3, even k included: more moduli, no gain. (iv) The 2-adic "impurity" of Part III has moved to 3: the condition $3 \nmid \beta$ produces $\epsilon_3(h)$ and $3 \mid |h|_3^2$, as $2 \nmid \beta$ produced $h_1 \equiv h_2 \pmod 2$. This is not an accident: the ramified prime is where the ideal of norm ℓ is principal, and removing it from the divisor sum of $r_D(\beta)$ is what makes the divisors $k < \sqrt{\beta}$ pair with β/k under χ. The clean level Dp matters for the holomorphic side (Section 3), not for the dispersion form; a level beyond 1/2 needs the bilinear saving over (m', k) of Part III, Section 4.5, in the Eisenstein setting as in the Gaussian one.

6. Outlook

The problem of Parts I–IV, $N = n + (N - n)$ with both summands norms from an imaginary quadratic field of class number one and few prime factors, is a binary additive problem of the Goldbach type with two sparse sequences, and it shares with Goldbach's problem the shape of its obstacles: a level of distribution 1/2 obtained from the modular side (here from a theta

series, there from the Bombieri–Vinogradov theorem), a sieve whose dimension (here 2) fixes the attainable pair (k, k') at that level, and a remainder beyond $1/2$ that is a bilinear form in Kloosterman sums with one argument fixed. What the present part adds to this picture is structural. The spectral support of the theta series is governed by the Atkin–Lehner signs at the ramified prime and at p (Theorem 3.3): the remainder sees only the newforms whose root number is $+1$ and whose signs match $\chi_{-D}(p)$, so the parity of the root number, which decides the vanishing of $L(1, \varphi)$, enters the sieve remainder at the level of individual forms; the Rankin–Selberg formula (Theorem 3.4) writes the remainder as a sum of products of two central values, a twist by χ_{-D} against the trivial one, so that the automorphic lift to GL_2 of the two quadratic fields (the one of the discriminant and the one of the twist) is where the arithmetic of N is encoded; and the norm, which measures the size of the remainder on average, is the reciprocal volume of $X_0(D)$ (Theorem 1.2(v)), the modular curve of the discriminant, which is rational for the four small discriminants $D \in \{3, 4, 7, 8\}$ and has genus $1, 1, 3, 5, 13$ for $D = 11, 19, 43, 67, 163$. We do not claim that any of this moves the Goldbach problem; the honest content is that the two problems have the same anatomy, and that in the theta-series setting every piece of that anatomy (main term, support, central values, norm) has been made explicit. The question that remains, here as there, is the bilinear form of Part III, Section 4.5.

Data and code

The scripts `subleading_log_anatomy.py` and `closed_form_B.py` (Section 2; they import the tables and the Siegel main term of Part II), `heegner_theta_unified.py` (Sections 3–4), `eisenstein_circle_check.py` (Lemma 5.1) and the plotting scripts `plot_scalloped_anatomy.py`, `plot_heegner_volume_scaling.py`, `plot_level77_duality.py`, with their outputs in `results/` and the working notes in `notes/`, are in the repository https://github.com/Ruqing1963/two-squares-part4. All computations are in floating point and are evidence, not proof; the theorems do not depend on them.

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